オウ　イシン
王　一臻

WANG Yizhen

Master | Mechanical Engineering | Hasegawa Lab, The University of Tokyo

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Profile

- Commercial CFD and design optimization. Multiphysics for aerospace propulsion (conjugate heat transfer, detailed-chemistry reacting flow, supercritical fluids) on HPC clusters, plus topology and parametric optimization, across ANSYS Fluent, OpenFOAM and ToffeeX.
- Master's thesis drove a turbulent pipe-flow rig by Bayesian optimization with no operator in the loop: 75% drag reduction, 40% power saving rate. JSAE Graduate Research Encouragement Award.
- In Japan on a work visa. Japanese with clients, English for research, Chinese first language.

Education

The University of Tokyo — Oct 2023 – Sep 2025

Master, Mechanical Engineering, Hasegawa Lab — Tokyo, Japan

- Automatic Optimization of Pulsating Waveform for Drag Reduction in Experiment of Turbulent Pipe Flow
- Built an unattended Bayesian-optimization loop over the rig, reaching **75% maximum drag reduction** and a **40% maximum power saving rate**

Thesis received the JSAE Graduate Research Encouragement Award

University of Melbourne — Jan 2019 – Dec 2019

Master, Mechanical Engineering (Withdrawn) — Melbourne, Australia

- Investigation of energetic motions in rough-wall turbulent boundary layers

Withdrew on receiving a fully funded PhD offer from the same university; enrollment was deferred under COVID-19 travel and visa restrictions and the offer lapsed in late 2022 under university policy

Shanghai Jiao Tong University — Sep 2014 – Sep 2018

Bachelor, Energy and Power Engineering; Minor in Musicology — Shanghai, China

- Dilution effects of DME and ethanol on methane flames
- Dynamic characteristics of hydrogen-cooled generator stator foundations

Work Experience

SOLIZE Partners — Apr 2026 – Present

CAE Analysis Engineer — Kanagawa, Japan

- Conjugate heat transfer, reacting-flow CFD with detailed chemical kinetics (Chemkin mechanisms in OpenFOAM), and supercritical-fluid analysis for aerospace propulsion, run in parallel on an HPC cluster
- Delivered optimization PoCs: piping support placement (heat loss, thermal stress, seismic natural frequency) and coupled fluid-thermal-structural with topology optimization
- Cross-validation across ANSYS Fluent / OpenFOAM / ToffeeX; built the internal OpenFOAM computing environment (Slurm)

X-NEW Education · OMEKA Education — 2020 – 2023

Part-time Instructor, AP / IB / A-Level Mathematics and Physics — Shanghai, China

- Taught mathematics and physics to students preparing for overseas university admission
- Began studying Japanese and prepared for graduate study in Japan, including approaching prospective supervisors

Skills

- **CFD and multiphysics:** ANSYS Fluent, OpenFOAM, ToffeeX. Conjugate heat transfer, reacting flow with detailed chemical kinetics (Chemkin), supercritical fluids, turbulence modeling.
- **Optimization:** Topology, parametric and hybrid optimization. Bayesian optimization with Gaussian-process surrogates, design of experiments.
- **Computing:** Python, C/C++, MATLAB. HPC clusters, Slurm, Git, GNU/Linux.
- **CAD and workflow:** SolidWorks, CATIA, SpaceClaim. Meshing, boundary conditions, post-processing, validation, technical reporting.

Languages: Mandarin Chinese (native) English (native level) Japanese JLPT N1 (business level)